

Domain Research Note

Title:

Bridging Generative AI and Optimization: A Collaborative Approach to Solving the Traveling Salesman Problem

Introduction:

Traveling Salesman Problem (TSP) is one of the most common optimization problems in the field of computer science. In TSP, there is a need to find the minimum length route that passes through some cities and ends at the starting city. This problem finds its applications in many real-world scenarios, including logistics and routing.

Primary Domain:

Generative AI for Optimization and Decision-Making

Problem Importance:

Efficiently solving the TSP is critical in industries where minimizing travel distance directly reduces cost, time, and fuel consumption. As the number of cities increases, the complexity of the problem grows exponentially, making it computationally challenging to solve using brute-force methods.

Proposed Approach:

In this study, a combined methodology will be applied to investigate the abilities of Generative AI to handle optimization problems. The research has been triggered by some studies on the application of Large Language Models (LLMs) in the optimization process, revealing both the strengths and weaknesses of AI in such a task.

A Large Language Model (LLM) will generate viable routes from natural language prompts, whereas Google OR-Tools will serve as a classical optimizer for finding optimal/near-optimal routes. The effectiveness of both techniques will be evaluated in order to discover the shortcomings of LLM optimization and the ways of compensating for them through classical optimization tools.

Dataset Used:

The dataset used for this project is sourced from Kaggle and contains multiple instances of TSP problems, including city coordinates. This allows testing the model on different problem sizes and evaluating consistency across multiple cases.

<https://www.kaggle.com/datasets/ziya07/traveling-salesman-problem-tsplib-dataset>

Objective:

This research aims to evaluate the feasibility of using Generative AI to solve combinatorial optimization problems and assess the efficiency of the approach compared to conventional approaches. The paper sheds light on the possible effectiveness of using AI reasoning and mathematical optimization methods together.